

TITLE PAGE

- **Problem Statement ID –** SIH26183
- **Problem Statement Title-** Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics
- **Theme-** BLOCKCHAIN & CYBERSECURITY
- **PS Category-** Software
- **Team ID-**
- **Team Name (Team Varanasi)**



Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Explanation of the Proposed Solution:

- Complaint Intake: Wallet addresses ingested from cybercrime complaint systems
- Automated Tracing: Blockchain transaction graph analysis, hop-by-hop
- VASP Identification: Clustering detects the exchange receiving funds
- Cross-Chain Detection: Flags intermediary wallets and cross-chain movement

How It Addresses the Problem :

- Speed: Manual tracing reduced to real-time analysis
- Multi-Chain Coverage: Supports tracing across multiple blockchain ecosystems
- Alert-Driven: Automated alerts on high-risk fund movement
- LEA-Ready: Standardized reports for law enforcement action

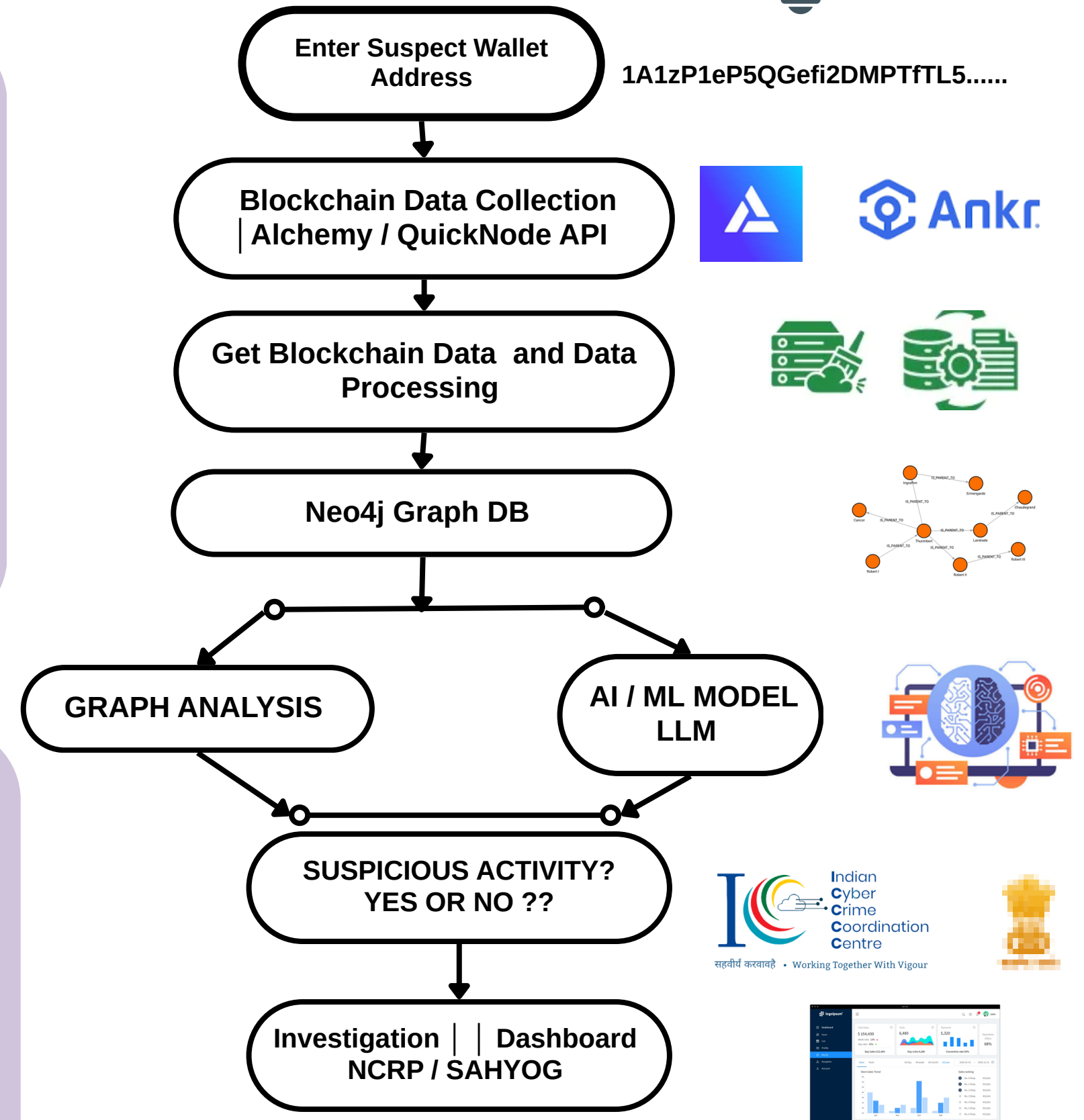
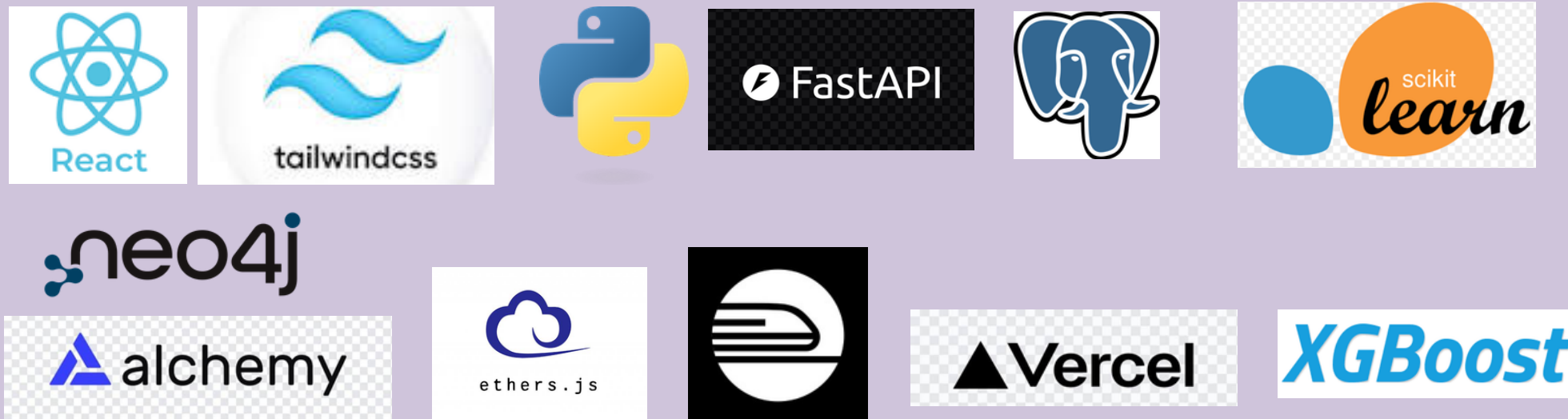
Innovation and Uniqueness of the Solution:

- Explainable Risk Scoring: AI/ML-assisted, with visible reasoning per finding
- Fund-Flow Visualization: Real-time analytics dashboard for investigators
- SAHYOG/NCRP Integration: Designed to connect with existing platforms
- Scalable Indexing: Built to support growing blockchain data volumes
- Pattern Recognition: Identifies recurring fraud typologies over time



TECH STACK :

- Frontend: React.js, Tailwind CSS, Cytoscape.js, Plotly
- Backend: Python, FastAPI, REST APIs, WebSockets
- Blockchain: Web3.py, Ethers.js, Alchemy/QuickNode, The Graph
- Data & Indexing: PostgreSQL, Custom Blockchain Indexer
- Graph Intelligence: Neo4j, NetworkX, GDS, Wallet Clustering
- AI/ML: Scikit-learn, XGBoost, PyTorch, Anomaly Detection & Risk Scoring
- Deployment: Nginx, Vercel, Railways, etc
- Integration: NCRP/SAHYOG APIs, Automated Investigation Reports



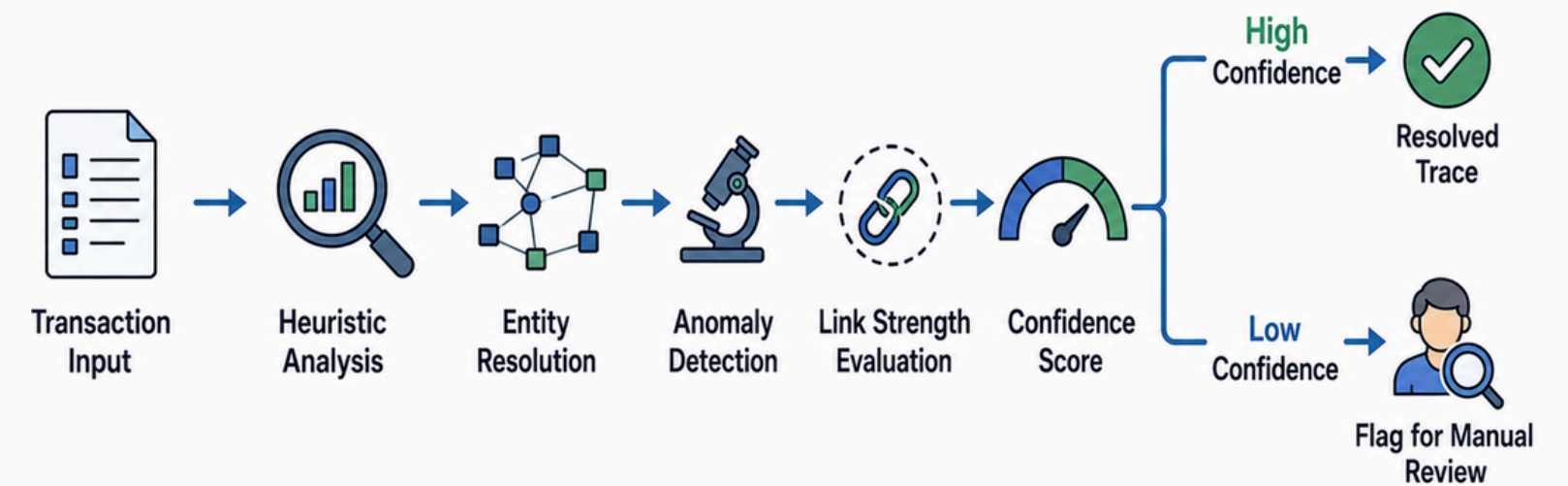
Technical feasibility :

- Uses publicly available blockchain transaction data.
- Supports blockchain APIs, nodes, graph analytics, and AI/ML.
- Can trace wallet-to-wallet and cross-chain fund movement.
- Can begin with major chains such as Bitcoin, Ethereum, Tron, and Polygon.

Operational feasibility :

- Accepts wallet addresses and transaction hashes from cybercrime complaints.
- Can support integration with NCRP, I4C, police systems, and authorized VASPs.

Strategies & Confidence-Scored Results



Phased Development Approach (MVP First)

PROCESS FEASIBILITY



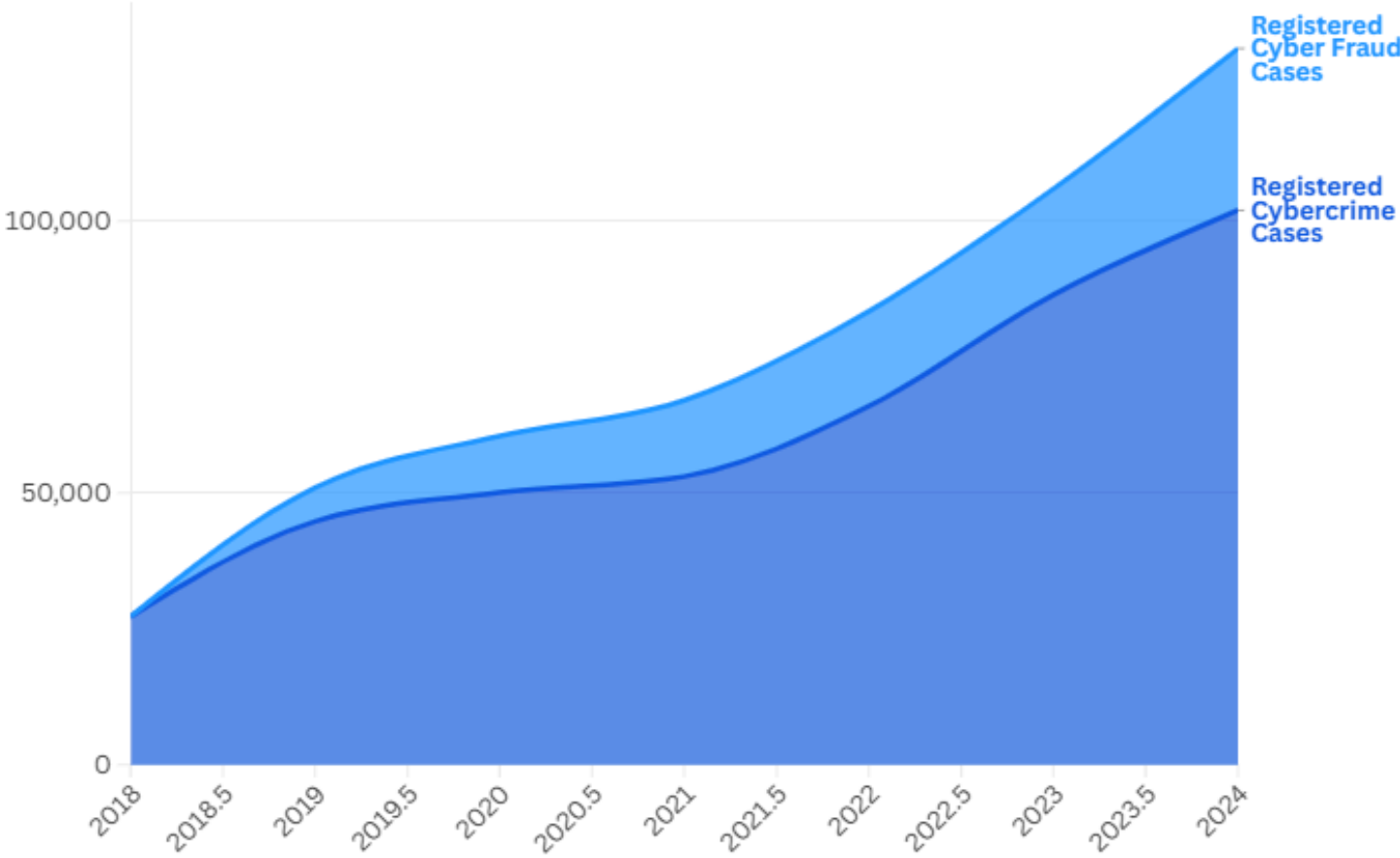
Economic viability :

- Reduces investigation time and analyst workload.
- Improves the possibility of early fund freezing and recovery.
- Can be implemented in phases, reducing initial development cost.

Expected impact :

- Faster identification of receiving exchanges or VASPs.
- Better tracing of fraud proceeds.
- Improved coordination between law-enforcement agencies and exchanges.

- **Faster investigations:** Automatically traces wallet transactions and reduces manual analysis time.
- **Quick identification of VASPs:** Helps identify the probable exchange, broker, or crypto service receiving fraud proceeds.
- **Improved fund recovery:** Early alerts can help investigators request preservation or freezing of funds more quickly.
- **Better evidence collection:** Generates transaction histories, wallet graphs, timestamps, and standardized reports.

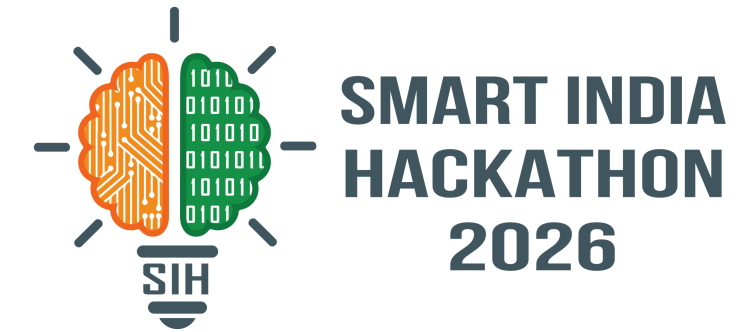


- Generates reliable, standardized, investigation-ready reports.
- Improves coordination between I4C, police, banks, and VASPs.
- Increases the possibility of recovering victims' funds.

The system enables faster crypto-fraud investigations by converting victim-reported wallet addresses into actionable intelligence for fund tracing, VASP identification, evidence collection, and asset recovery.

RESEARCH AND REFERENCES

Team
Varanasi



NIST – Blockchain Technology(<https://www.nist.gov/blockchain>)

- Studied blockchain fundamentals, architecture and distributed ledger technology.
- Understanding of blockchain transactions and traceability.

arXiv – Blockchain Technology(<https://arxiv.org/html/2505.08772v1>)

- Studied blockchain architecture, cryptography and consensus mechanisms.
- Understanding of scalability and cross-chain interoperability.

ICFACT 2025 – Blockchain Forensics(<https://www.atlantis-press.com/proceedings/icfact-25/126023943>)

- Studied cryptocurrency transaction analysis and blockchain forensics.
- Address clustering, network mapping and suspicious transaction detection.
- Inspired graph-based wallet and fund-flow analysis.

I4C – National Cyber Crime Reporting Portal (<https://i4c.mha.gov.in/ncrp.aspx>)

- Studied the existing cybercrime reporting and investigation ecosystem.
- Understanding of LEA coordination and cyber-fraud reporting workflows.
- Inspired integration with existing law-enforcement systems.